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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, DESIGN AND MANUFACTURING KANCHEEPURAM

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PROTOTYPING AND TESTING - CODE : (DS3001)

SMART WASTE SEGREGATOR

A Project Report

submitted by

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DECLARATION OF ORIGINALITY

We, **BATTU RAKESH (EC21B1025), NARIBOINA BHANUPRAKASH (EC21B1039), R SIVA KIRAN (EC21B1045), PATAN IRFANKHAN (EC21B1056) –** ***(Team - F18)*** hereby declare that the material presented in the Project Report titled **PROTOTYPING AND TESTING - CODE : (DS3001) SMART WASTE SEGREGATOR** represents original work carried out by me in the **Department of Electronics and Communication Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.

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ABSTRACT

Management of solid waste is essential to our health and environmental protection. Separation of the wastes in order to reduce, re-use, recycle, and recover energy from them or to process them for further use, is key to the proper management of the waste. . In order to sort the wastes, a sorting machine is designed . The machine is designed with the major components being the slow moving conveyor, shafts, electric motor,fan and many other sensors.Based on the theoretical calculations the segregator is capable of sorting different types of wastes like metallic(ferrous and non ferrous),wet waste and dry waste(plastics and paper).the smart waste segregator is capable of communication with the user to share the bin capacity status making it easy for the user to replace the filled bins.

TABLE OF CONTENTS

ABSTRACT **i**

1 SMART WASTE SEGRAGATOR **1**

1.1 Introduction 1
1.2 Literature Survey 2
1.3 Suggested Solution 3
1.4 Methodology 4
1.4.1 Design Theory and Working 4
1.4.2 Circuit connections 6
1.5 Billing of Materials 6
1.6 Roles and Responsibilities 7
1.7 Future Scope 7
1.8 Conclusion 8

CHAPTER 1

SMART WASTE SEGRAGATOR

1.1 Introduction

In today's rapidly urbanizing world, waste segregation has become a monumental challenge for communities worldwide. As populations burgeon and urban areas expand, the management of waste has become increasingly complex, posing significant environmental, health, and social concerns. Amidst this backdrop, the need for innovative solutions to streamline waste segregation processes has never been more pressing.Waste generation and disposal is naturally part of any developing and industrial society. The problem of solid waste is therefore a universal one as waste exists in every society and the wastes from both commercial and domestic sources are considerably growing every day.Waste management consists of the whole set of activities related to the handling, treating, disposing or recycling the waste materials. Solid wastes not properly handled and disposed into open spaces poses danger to human health as well as the environment. Waste management is concerned with the generation, prevention, monitoring, characterization, treatment, reuse and residual disposition of solid wastes.Waste generation increasing beyond the acceptable levels led to the changes in the framework directives and policies which formulated waste management hierarchy. In waste management hierarchy, the waste should follow "reduce, reuse, recycle and recover energy" before it is dumped into the landfill.

The general composition of daily household waste consists of paper, plastics, metallic waste, organic waste, glass etc.Metallic waste consists of ferrous and non ferrous metallic fractions, these waste therefore need to be seperated in order to be able to reuse, recycle, reduce, renew them or can be used for generating fuel, so in recent times there are emphasis on the waste sorting before recycling and reusing which helps to reduce waste mangement cost.

This report explores one such solution: the implementation of smart waste segregators. By leveraging technology to enhance waste sorting efficiency and address the mounting

challenges associated with waste management.

1.2 Literature Survey

Global municipal solid waste generation is expected to have reached 2.02 billion tons in 2005–06, an increase of 7 percent annually since 2003. In order to allow waste to be routed straight for processing, this study suggests an Automated Waste Segregator (AWS), which is an inexpensive and simple-to-use solution for a domestic waste segregation system. Its purpose is to separate the waste into two categories: dry and wet. Capacitive sensors are used by the AWS to differentiate between wet and dry trash. The outcomes of the tests demonstrate that the AWS has been successfully used to separate waste into dry and moist waste. [1]

In [2] the main objective of this project is to design a system using Arduino Mega for automatic segregating of waste at source and capable of cleaning. IR sensor estimates the distance and the status of the bin will be send through IFTTT (If this then that) for sending SMS. This bin can be used at places like offices, apartments, shopping malls etc. This system will be useful in making Waste Management in smart cities automated without the human intervention.

In [3] they mainly focused on to segregate metallic,wet and dry waste using various sensors and Arduino,metal sensor and moisture sensor.

In [4] their focus is on a separate waste at the houses in the society into dry waste and wet waste using a relay circuit driven by a comparator circuit. This waste is collected at the container. The container has a sonar sensor used to detect the level of the garbage collected into it.

In [5] paper it shows how two conveyer belts are used to seperate various waste materials including ferrous material but it has large area and not so easy to make and not so cost effective and this has taken reference form [6]

Advanced technologies used in industries

advancements in technology have led to the development of automated waste segregating machines equipped with various sensors, sorting mechanisms, and artificial intelligence algorithms.

- One common technology used in waste segregating machines is optical sorting. Optical sensors can identify different types of materials based on their optical properties such as color, shape, and reflectivity. This allows the machine to sort materials like plastic, glass, paper, and metal with high accuracy and speed.
- Another technology employed in waste segregating machines is near-infrared (NIR) spectroscopy. NIR sensors can analyze the molecular composition of materials, enabling the identification and sorting of different types of plastics and other materials based on their chemical signatures.
- Robotic arms and mechanical sieves are utilized in some advanced waste segregating machines to physically separate different materials based on size, shape, and density.

But all these methods are very expensive and complex systems which are very hard to build and operate making them less suitable options for household usage.

1.3 Suggested Solution

In addressing the challenges outlined in the preceding sections, it becomes evident that a comprehensive solution is imperative to solve the issues and achieve our desired objectives.

We intend to create a miniaturized household smart waste segregator which is cost efficient and can be easily used in the houses or in a municipal locality.

Use of cv and ML trained systems increases the cost of manufacturing also need's a large setup or working unit and is difficult to operate,so we intend to use the material properties like density,metallic properties and other properties with the help of conveyer belt to segregate different types of waste.

1.4 Methodology

1.4.1 Design Theory and Working

The following parts were considered for the design of smart waste segregation machine:

- Inductive proximity sensor
- Capacitive proximity sensor
- IR sensor
- Ultrasonic sensor
- GSM module
- Servo motor
- DC motor for conveyor belt

Design Specifications for Conveyor

The following are the design specifications for the horizontal flat conveyor:

- In order to accommodate a reasonable quantity of waste per unit time, a belt width of 300mm is considered.
- A length of 50cm is considered for the slow moving conveyor belt.

Working

- The waste enters the system and is detected by a sensor.
- The system then checks if the trash bin is empty. If it is, the system starts sorting the waste.
- The first stage of sorting involves the use of an inductive proximity sensor to see if the trash is metal.
- If the trash is metal, the metallic bin opens.
- If the trash is not metal, the system moves on to the next stage of sorting.
- The next stage of sorting involves a moisture sensor. The system checks to see if the trash is wet.
- If the trash is wet, the wet bin opens.
- If the trash is not wet, the system moves on to the next stage of sorting.
- The final stage of sorting involves a capacitive sensor to see if the trash is plastic.

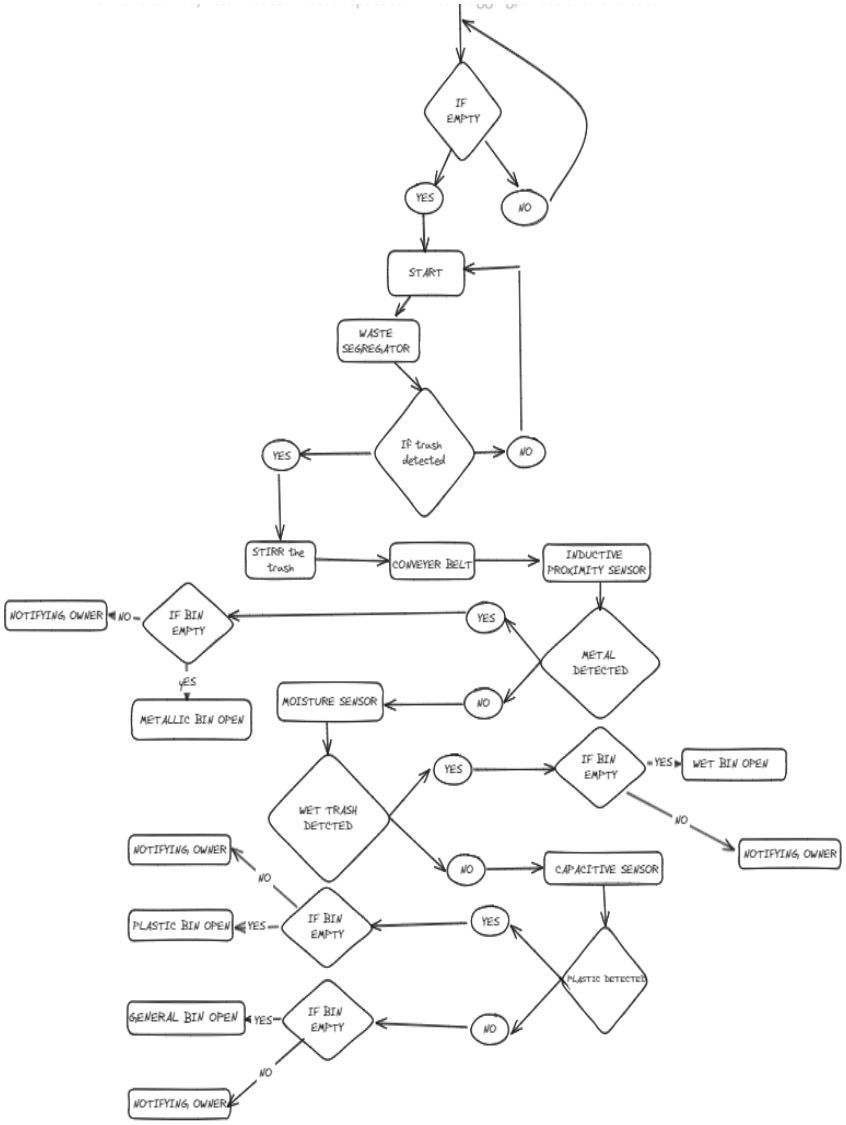


Figure 1.1: flow chart working of the smart waste segregator

- If the trash is plastic, the plastic bin opens.
- If the trash is not plastic, it is categorized as general waste and is deposited in the general bin.
- If the system is unable to sort the trash or the bin is full, the owner is notified.

This is a basic methodology/process used to sort different types of waste waste.

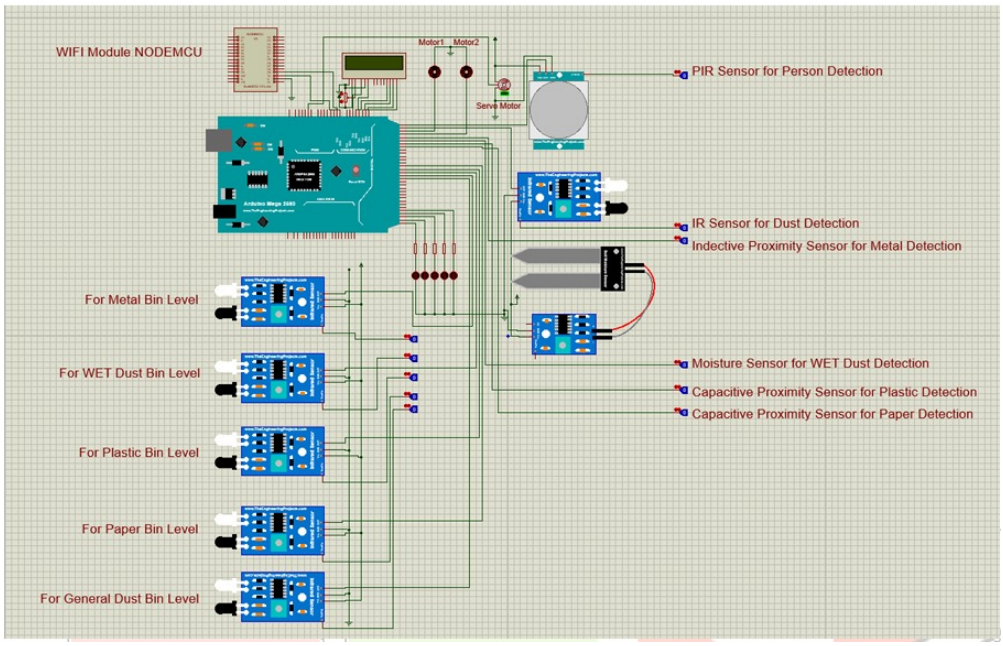


Figure 1.2: circuit connection of arduino and all sensors

1.4.2 Circuit connections

1.5 Billing of Materials

Billing of all the sensors

Component	Quantity	Cost
Inductive proximity sensor	1	Rs.250
Capacitive proximity sensor	1	Rs.530
IR sensor	3	Rs.78
Ultrasonic sensor	1	Rs.99
GSM module	1	Rs.239
moisture sensor	1	Rs.36
DC motor	1	Rs.476
Servo motor	2	Rs.470
Total	11	Rs.2178

- Outer body of the bin
- different bins for separated waste
- few installation materials

1.6 Roles and Responsibilities

Siva Kiran : Studied and working on the circuit designing and integration of various sensors to work together,helped in deciding the problem statement and proposed a solution also worked on making the presentation.

Rakesh : Focused on researching various articles related to the problem statement and finding the gaps in them and working on Outer body design and hardware installations and circuit implementations, also work on making the report.

Irfankhan : Researched on the solutions to the gap that is there in the existing solution to the problem statement and proposed sensors that can be used in the addition to the existing solution and also worked on making the report and will be working on the sensor implementation using arduino and the coding part .

Bhanuprakash : Also working on circuit and the sensor requirement part,helped in making the circuit simulations,worked on the basic idea of the bin and also helped in preparing the presentation.

1.7 Future Scope

As we conclude our analysis and recommendations for the present project, it's crucial to look ahead and consider the potential for further advancements and improvements of this project.

There are few other categories in which the waste can further be segregated which involves further differentiation of metallic waste and differentiation between different types of plastics.

1.8 Conclusion

In this report, we have studied various methodologies for proper waste segregation.The study shows how mainly IoT and CV techniques are used for the waste segregation.The project when implemented will help to create more resources for recycling as it decreases the probability of contamination, hence increasing the resources for recycling.

REFERENCES

[1] S. B. V. S. M. Bhoomika P, Sonika, "Segregation of waste-a survey," *International Research Journal of Engineering and Technology (IRJET)*, 2020.

[2] S. S. Sidhanth Pandey, Sairam Makkena, "Smart garbage segregator using iot," *INTERNATIONAL JOURNAL OF CREATIVE RESEARCH THOUGHTS - IJCRT (IJCRT.ORG)*, vol. 9, 2021.

[3] S. V. T. S. P. Ajinkya Vyavhare, Lokesh Jawale, "Automatic waste segregator," *IJARIIE.com*, vol. 8, 2022.

[4] H. S. K. P. K. R. P. Chethan Siddeshwar K R, Gunasagar C M, "Automatic waste segregation system," 2021.

[5] S. K. Z. Olawale James OKEGBILE, Nicholas Akhaze MUSA, "Design and fabrication of a solid waste sorting machine," *ABUAD Journal of Engineering Research and Development (AJERD)*, vol. 1, 2018.

[6] P. U. . Imoh, E. U. Emmanuel, "Domestic solid waste management in a rapidly growing nigerian city of uyo," *http://www.krepublishers.com/02 journals...9-11-2094-Ukpong-1-E-Tt.pdf*.